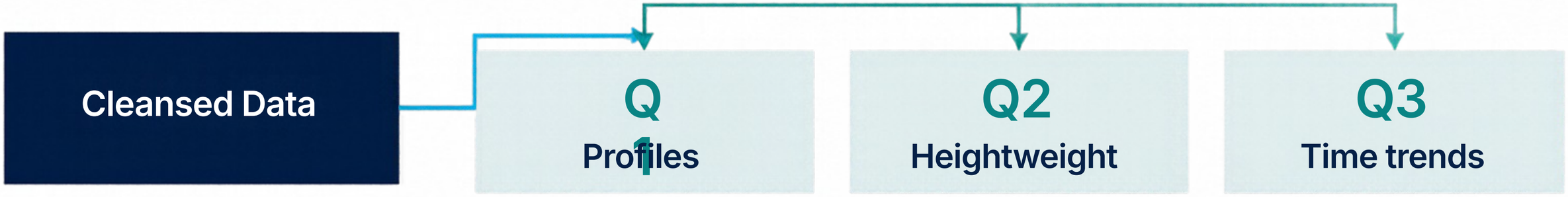
Our project asks three connected questions

Our project studies the physical profiles of Indian Summer Olympic athletes.

We ask three questions: how do profiles differ across sports, does the heightweight relationship vary across sports after accounting for sex, and how does weight change over Olympic years after adjustment?

Our workflow starts with cleaning, moves to descriptive comparisons, and then uses regression to examine these relationships.





Project reports and notebooks, Group 13.

30 seconds

We prepared a consistent dataset

We used the 120 Years of Olympic History dataset and filtered it to Indian Summer Olympic entries.

We removed duplicates and records missing height or weight.

We filled missing ages using sport-and-sex medians, with an overall median fallback, and standardized labels.

The cleaned data contain 835 event-level records covering 522 athletes from 1936 to 2016.

We then assigned sports and selected events to seven functional archetypes.

83
event records

5

52
unique athletes

2

1936–2016

Combat, Endurance, Precision, Racket Sports, Speed & Power, Team Sports, Technical / Mixed.

cleaning notebooks; Question 1 report.

Q1: We compared athlete profiles

For Question 1, we retained one record per athlete, year and archetype, giving 737 observations.

We compared age, height, weight and BMI using summary statistics and distribution plots, including comparisons by sex and Athletics sub-discipline.

Precision had the highest mean age, Racket Sports the greatest mean height, and Endurance the lowest mean weight and BMI.

These describe group differences without establishing their causes.



Precision	Racket Sports	Endurance
29.21	176.66	61.04
years	cm	kg

Q2: We improved the heightweight model

We first fitted a height-only regression using 732 athlete-year records.

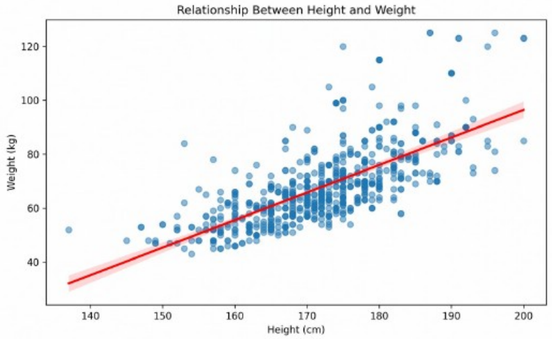
Height and weight were positively correlated, but the residuals differed across archetypes.

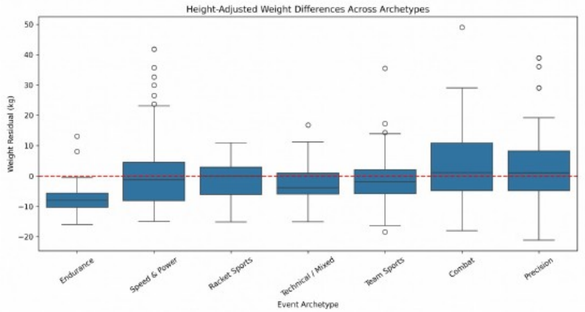
We then added sex, archetype and height-by-archetype interactions, allowing different height slopes.

We excluded five records belonging to multiple archetypes.

On the same 727 records, R-squared increased from about 0.487 to 0.576.

An interaction F-test supported different slopes under ordinary OLS assumptions.





On the same 727 records, R^2 increased from 0.487 to 0.576 after adding sex, archetype and height-by-archetype interactions.

Interaction F-test: $p \approx 6.01 \times 10^{-10}$

question2-report.pdf; Task2regression-analysis_final.ipynb.

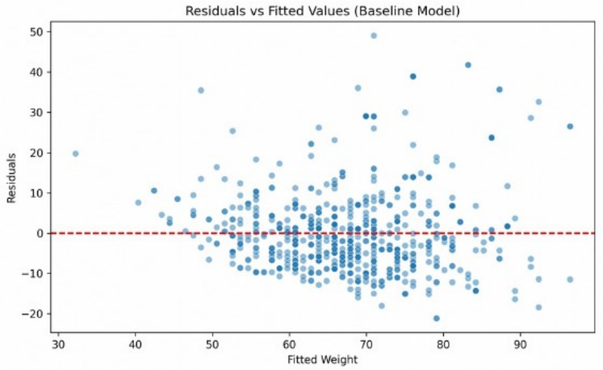
We checked errors and prediction uncertainty

Next, we inspected residual and Q–Q plots to check model assumptions.

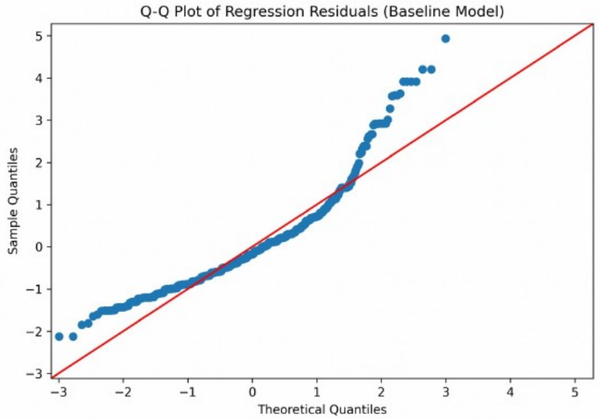
We split athletes, rather than individual rows, into roughly eighty percent training and twenty percent testing, so the same athlete never appeared in both sets.

We also calculated individual 95 percent prediction intervals.

Their width showed that predicting one athlete's weight remains uncertain, even when the model explains more overall variation.



Residuals vs fitted values



Baseline-model Q–Q plot

Departures in the tails highlight limits to the normal-error assumption.

question2-report.pdf.

Q3: We examined weight changes over time

Question 3 used the 835 event-level records.

We compared a model containing height and Olympic year with one that also included sex and archetype.

The estimated year association increased from 0.71 to 1.00 kilograms per decade after adjustment.

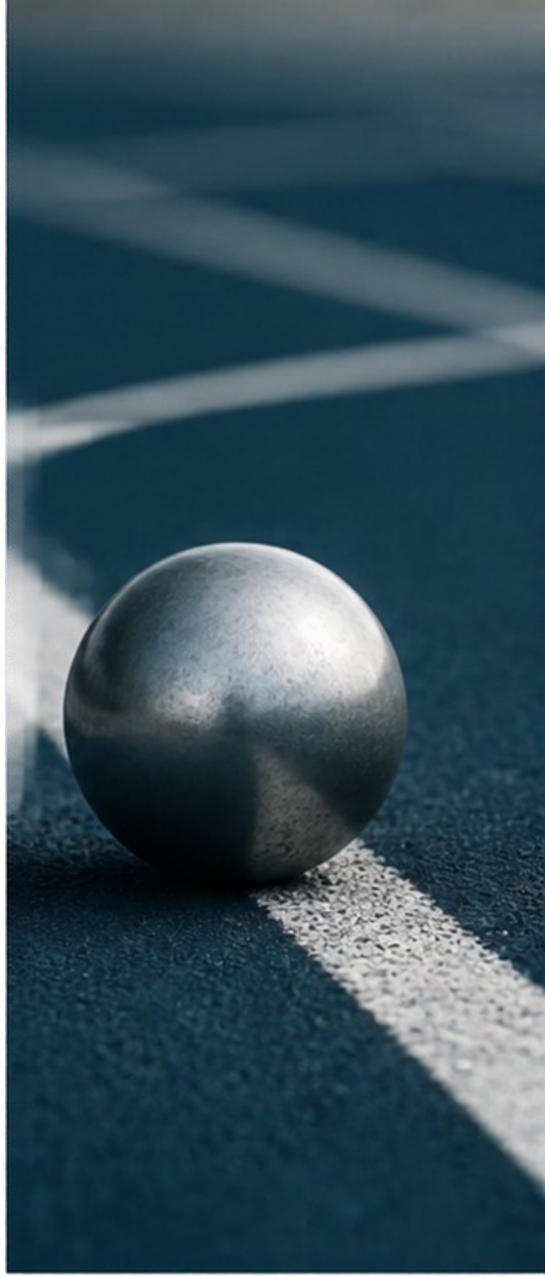
We used standard errors clustered by athlete and checked men's hockey separately, where the estimate was 1.31 kilograms per decade.

These are associations across editions, not individual athletes gaining weight over time.

Height + year:	0.71 kg/decade
+ sex + archetype:	1.00 kg/decade
Men's hockey check:	1.3 kg/decade

1

Hockey uses 198 records from 132 athletes.
Q3 retains event-level weighting.



Untitled1.ipynb saved model outputs.

What the full workflow tells us

Our workflow connects data cleaning, descriptive comparisons, regression and model checks.

The results suggest that physical profiles and height-weight associations vary across sporting groups, while adjusted weight also shows a positive association with Olympic year.

The main limitations are missing-measurement filtering, unequal group sizes, sex imbalance and repeated athlete records.

Our conclusions describe this recorded sample and do not establish causal effects.

Sporting context matters when interpreting athlete measurements.

